

- 1 Fig. 1.1 shows a 1000 N uniform thin rod being towed by a force T and moving at constant horizontal velocity.

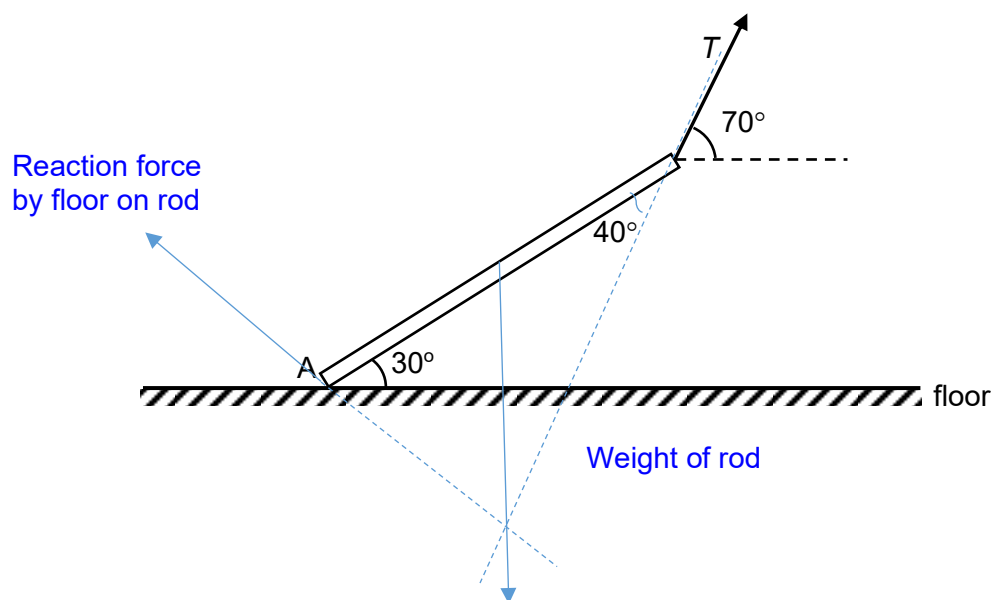


Fig. 1.1

- (a) State the conditions required for a body to be in *equilibrium*.

A body is in equilibrium if the net force in any direction on the body is zero and the net torque about any axis on the body is zero.

.....[2]

- (b) On Fig. 1.1, draw and label the **two** other forces acting on the rod.

[2]

- (c) Given angle θ is 70° , determine force T .

Let L be the length of the rod.

Taking moment about A,

$$\text{clockwise moment due to weight} = 1000\left(\frac{L}{2}\cos 30^\circ\right) \quad [1]$$

$$\text{anticlockwise moment due to force } T = T(L\sin 40^\circ) \quad [1]$$

At equilibrium net moment is zero,

$$1000\left(\frac{L}{2}\cos 30^\circ\right) = T(L\sin 40^\circ)$$

$$T = 670 \text{ N} \quad [1]$$

force $T = \dots\dots\dots \text{ N} \quad [3]$

2. The International Space Station (ISS) orbits the Earth at a height of 4.1×10^5 m above the Earth's